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AA-CLIP: Enhancing Zero-Shot Anomaly Detection via Anomaly-Aware CLIP

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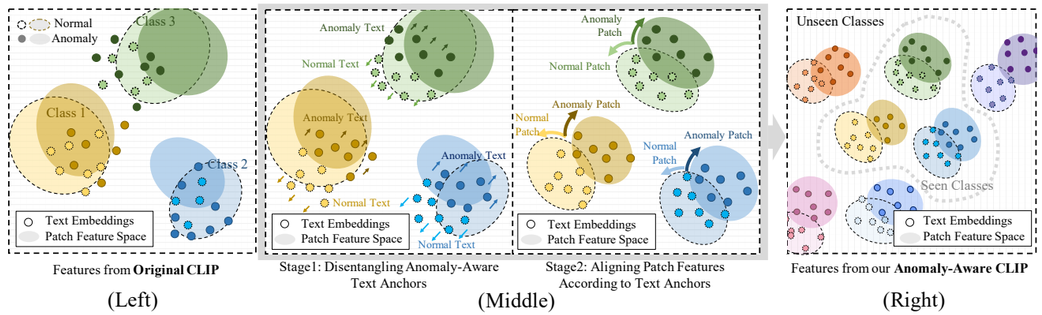


Figure 1. **(Left) CLIP's anomaly unawareness:** Category-level image-text alignment in pre-training leads to CLIP's vague distinctions in anomaly/normal semantics and inaccurate patch-text alignment. **(Middle) Our two-stage adaptation strategy:** In Stage1, anomaly and normal text features are disentangled as anchors in text space; in Stage2, patch-level visual features are trained to align to these anchors, forming Anomaly-Aware CLIP. **(Right) Generalizable anomaly awareness:** Our method enables CLIP with generalizable anomaly awareness for both known and unseen classes.

Abstract

Anomaly detection (AD) identifies outliers for applications like defect and lesion detection. While CLIP shows promise for zero-shot AD tasks due to its strong generalization capabilities, its inherent ***Anomaly-Unawareness*** *leads to limited discrimination between normal and abnormal features. To address this problem, we propose* ***Anomaly-Aware CLIP*** *(AA-CLIP), which enhances CLIP's anomaly discrimination ability in both text and visual spaces while preserving its generalization capability. AA-CLIP is achieved through a straightforward yet effective two-stage approach: it first creates anomaly-aware text anchors to differentiate normal and abnormal semantics clearly, then aligns patch-level visual features with these anchors for precise anomaly localization. This two-stage strategy, with the help of residual adapters, gradually adapts CLIP in a controlled manner, achieving effective AD while maintaining CLIP's class knowledge. Extensive experiments validate AA-CLIP as a resource-efficient solution for zero-shot AD tasks, achieving state-of-the-art results in industrial and medical applications. The code is available at* https://github.com/Mwxinnn/AA-CLIP.

1. Introduction

Anomaly detection (AD) involves modeling the distribution of a dataset to identify outliers, such as defects in industrial products [2] or lesions in medical images [13]. Despite that previous AD frameworks [10, 11, 15, 23, 31, 58] effectively detect anomalies when sufficient labeled data is available for specific classes, their high resource demands often limit their generalization ability to novel and rare classes. This limitation is particularly challenging in real-world scenarios where collecting comprehensive labeled datasets for AD is often infeasible, necessitating the exploration of low-shot

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AA-CLIP：通过异常感知 CLIP 提升零样本异常检测

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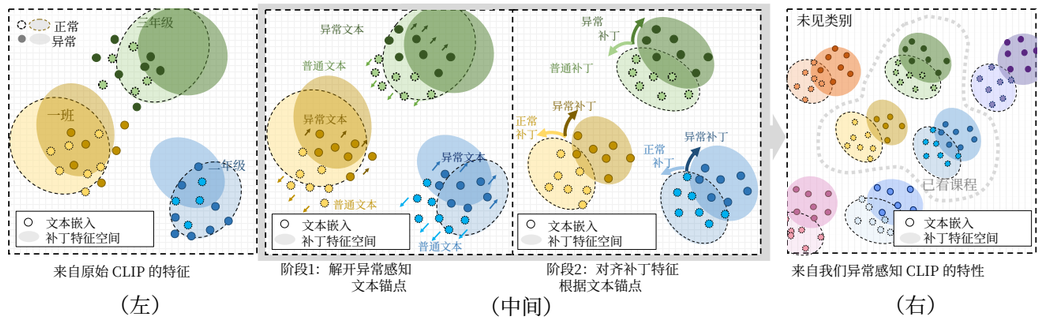


图 1。（左）CLIP 的异常无感知性：预训练中的类别级图文对齐导致 CLIP 在异常/正常语义上区分模糊，并且局部图像与文本的对齐不准确。（中）我们的两阶段适应策略：在阶段 1 中，异常和正常文本特征在文本空间中被解耦为锚点；在阶段 2 中，局部视觉特征被训练以对齐这些锚点，从而形成异常感知 CLIP。（右）可推广的异常感知性：我们的方法使 CLIP 对已知和未知类别都具备可推广的异常感知能力。

摘要

异常检测（*AD*）用于识别异常值，应用于缺陷和病变检测等场景。尽管*CLIP*因其强大的泛化能力在零样本*AD*任务中表现出潜力，但其固有的异常不敏感性导致对正常与异常特征的区分能力有限。为了解决这一问题，我们提出了异常感知*CLIP*（*AA-CLIP*），该方法在保留*CLIP*泛化能力的同时，增强了其在文本和视觉空间中的异常区分能力。*AA-CLIP*通过一个简单却有效的两阶段方法实现：首先创建异常感知文本锚点，以清晰区分正常和异常语义，然后将分块级视觉特征与这些锚点对齐，以实现精确的异常定位。在残差适配器的帮助下，这一两阶段策略逐步以受控方式对*CLIP*进行适配。

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ner，在保持 *CLIP* 类别知识的同时，实现有效的 *AD*。大量实验验证了 *AA-CLIP* 作为零样本 *AD* 任务的资源高效解决方案，在工业和医疗应用中取得了最先进的结果。代码可在 *https://github.com/Mwxinnn/AA-CLIP* 获取。

1. 引言

异常检测（AD）涉及对数据集的分布进行建模以识别异常值，例如工业产品中的缺陷 [2]或医学影像中的病变 [13]。尽管之前的AD框架 [10, 11, 15, 23, 31, 58] 在特定类别有足够标注数据时能够有效检测异常，其高资源需求通常限制了其对新颖和稀有类别的泛化能力。这个限制在现实场景中尤为具有挑战性，因为为AD收集全面的标注数据集通常不可行，因此需要探索低样本学习。

learning and transfer learning approaches.

Contrastive Language-Image Pretraining (CLIP) model has emerged as a promising solution, demonstrating remarkable generalization capabilities across various zero-shot tasks [24–26, 42]. Building upon CLIP's success, several recent studies have adapted CLIP for few/zero-shot AD tasks by utilizing anomaly-related descriptions to guide the detection of anomalous regions. Specifically, the vision encoder is trained to map anomaly images to visual features that align more closely with text features of abnormal descriptions than with those of normal descriptions [29, 30, 49, 60]. Further works [6, 7, 17, 41] have focused on enhancing CLIP's patch-level feature representations to achieve better alignment with text features, resulting in improved anomaly localization performance.

These methods depend on text features that need to be anomaly-aware to effectively differentiate abnormalities. However, recent studies highlight CLIP's limitations in fine-grained semantic perception and reasoning [21, 22, 36, 38, 45, 46]. Upon exploring CLIP's texture features for AD, we observe that while CLIP's text encoder effectively captures object-level information, it struggles to reliably distinguish between normal and abnormal semantics. As shown in conceptual visualization Fig. 1(left) and sampled examples in Fig. 2, CLIP has the intrinsic **Anomaly-Unawareness** problem: the overlap of normal and abnormal texture features hampers the precision of text-guided anomaly detection. We argue that making CLIP anomaly-aware — by establishing clearer distinctions between normal and abnormal semantics in the text space — is essential for guiding the vision encoder to precisely detect and localize anomalies.

This observation drives us to improve CLIP-based zero-shot AD through enhancing anomaly discrimination in text space, achieved with our method **Anomaly-Aware CLIP (AA-CLIP)** — a CLIP model with anomaly-aware information encoded. AA-CLIP is implemented through a novel two-stage adaptation approach. In the first stage, AA-CLIP adapts the text encoder with frozen visual encoder, creating "anchors" for anomaly-aware semantics within the text space for each trained class. As illustrated in Fig. 1(middle), each class's text features are disentangled to distinct anchors, with clear abnormality discrimination. Notably, this disentanglement also applies to novel, unseen classes, supporting effective zero-shot inference in AD tasks (refer to Fig. 1(right)). In the second stage, AA-CLIP aligns patch-level visual features with these specially adapted texture anchors, guiding CLIP's visual encoder to concentrate on anomaly-relevant regions. This two-stage approach ensures a focused and precise anomaly detection framework.

Importantly, as CLIP is extensively trained on massive data, to preserve its pre-trained knowledge, we utilize simple-structured Residual Adapters in both stages. This design enables a controlled adaptation of CLIP while enhancing its capability to handle fine-grained AD tasks without sacrificing its generalization ability.

Our extensive experiments in both industrial and medical domains demonstrate that our straightforward approach equips CLIP with improved zero-shot AD ability, even in data-limited scenarios. By training with a minimal sample — such as one normal sample and one anomaly sample (2-shot) per class — and testing across unseen datasets, our method achieves zero-shot performance comparable to other CLIP-based AD techniques. With only 64-shot of each class seen in the training set, our method reaches state-of-the-art (SOTA) results in cross-dataset zero-shot testing, validating our method's ability to maximize the CLIP's potential for AD with a minimal data requirement.

Our contributions are summarized as follows:

1. *Anomaly-Aware CLIP with enhanced and generalizable anomaly-discriminative ability.* We introduce AA-CLIP which is more sensitive to anomalies sequentially in text and visual spaces, encoding anomaly-aware information into the original CLIP.
2. *Efficient adaptation using residual adapters.* We implement simple residual adapters to boost zero-shot anomaly detection performance without compromising the model's generalization ability.
3. *SOTA performance with high training efficiency.* Our method achieves SOTA results across diverse datasets, showing robust anomaly detection capabilities even with limited training samples.

2. Related Work

Traditional Anomaly Detection in images involves modeling the normal data distribution to detect rare and diverse unexpected signals within visual data [44, 52, 53, 61]. Reconstruction-based [11, 16, 32, 33, 54, 56], augmentation-based [31, 44, 48, 55, 58] and discriminative [10, 15, 23, 31, 43, 61] methods are typically used to facilitate better modeling. Despite the huge progress of traditional anomaly detection methods, their effectiveness relies heavily on a well-modeled normal data distribution. Without sufficient normal data, their ability to accurately detect anomalies is significantly reduced.

CLIP, trained on a vast amount of image-text data, leverages contrastive learning alongside powerful language models and visual feature encoders to capture robust concepts. This combination enables CLIP to achieve impressive zero-shot performance on image classification, as it can generalize well to new categories without requiring task-specific training [24–27, 42, 51]. More recently, numerous studies [9, 12, 35, 39] have explored ways to transfer the knowledge embedded in CLIP models to a variety of downstream tasks, yielding promising results in fields like image captioning, image-text retrieval, and image generation. These efforts demonstrate CLIP's versatility and potential to drive

学习和迁移学习方法

对比语言-图像预训练（CLIP）模型已成为一种有前景的解决方案，在各种零样本任务中展示了显著的泛化能力 [24–26, 42]。在CLIP成功的基础上，近年来的几项研究通过利用与异常相关的描述来指导异常区域的检测，将CLIP适应于少样本/零样本异常检测任务。具体来说，视觉编码器被训练以将异常图像映射到与异常描述文本特征比与正常描述文本特征更紧密对齐的视觉特征 [29, 30, 49, 60]。进一步的工作 [6, 7, 17, 41] 则专注于增强CLIP的块级特征表示，以实现与文本特征的更好对齐，从而提高异常定位性能。

这些方法依赖于需要具备异常感知能力的文本特征，以有效区分异常。然而，最近的研究指出了CLIP在细粒度语义感知和推理方面的局限性 [21, 22, 36, 38,45, 46]。在探索CLIP的纹理特征用于异常检测（AD）时，我们观察到，虽然CLIP的文本编码器能够有效捕捉对象级别的信息，但其在可靠区分正常与异常语义方面表现不佳。如概念可视化图1（左）和图2中的示例所示，CLIP存在内在的异常无感问题：正常与异常的纹理特征重叠，阻碍了文本引导的异常检测的精确性。我们认为，使CLIP具备异常感知能力——通过在文本空间中建立正常与异常语义的更清晰区分——对于引导视觉编码器精确检测和定位异常是必不可少的。

这一观察促使我们通过增强文本空间中的异常识别来改进基于CLIP的零样本异常检测（AD），这一目标通过我们的方法异常感知CLIP（AA-CLIP）实现——一个编码了异常感知信息的CLIP模型。AA-CLIP通过一种新颖的两阶段适配方法实现。在第一阶段，AA-CLIP在视觉编码器冻结的情况下适配文本编码器，为每个训练类别在文本空间中创建异常感知语义的“锚点”。如图1（中部）所示，每个类别的文本特征被解耦到不同的锚点，具有明显的异常性区分。值得注意的是，这种解耦同样适用于新颖的、未见过的类别，从而支持在异常检测任务中有效的零样本推理（参见图1（右））。在第二阶段，AA-CLIP将补丁级视觉特征与这些经过特殊适配的文本锚点对齐，引导CLIP的视觉编码器专注于与异常相关的区域。这种两阶段方法确保了集中和p

重要的是，由于CLIP在大量数据上进行了广泛训练，为了保留其预训练知识，我们在两个阶段都使用了结构简单的残差适配器。这个设计使得CLIP能够被控制性地适配，同时增强其处理细粒度异常检测任务的能力，同时不牺牲其泛化能力。

我们在工业和医疗领域的大量实验表明，我们的简单方法能够赋予 CLIP 更强的零样本异常检测（AD）能力，即使在数据有限的情况下也是如此。通过使用最少样本进行训练——例如每个类别仅一个正常样本和一个异常样本（2-shot）——并在未见过的数据集上进行测试，我们的方法在零样本性能上达到了与其他基于 CLIP 的异常检测技术相当的水平。在训练集中每个类别仅看到 64 个样本的情况下，我们的方法在跨数据集零样本测试中达到了最新的（SOTA）结果，从而验证了我们的方法能够在最少数据需求下最大化 CLIP 在异常检测中的潜力。

我们的贡献总结如下：

1. 异常感知的 CLIP，具有增强的、可泛化的异常区分能力。我们引入了 AA-CLIP，它在文本和视觉空间中对异常序列更敏感，将异常感知信息编码到原始 CLIP 中。 2. 使用残差适配器进行高效适配。我们实现了简单的残差适配器，以提升零样本异常检测性能，同时不影响模型的泛化能力。

3. 具有高训练效率的SOTA性能。我们的方法在各种数据集上都取得了SOTA结果，即使在训练样本有限的情况下，也显示出稳健的异常检测能力。

2. 相关工作

图像中的传统异常检测涉及对正常数据分布进行建模，以检测视觉数据中稀有且多样的意外信号 [44, 52, 53, 61]。基于重建的方法[11, 16, 32, 33, 54, 56]、基于增强的方法 [31, 44, 48, 55, 58] 和判别方法 [10, 15, 23, 31, 43, 61] 通常用于促进更好的建模。尽管传统异常检测方法取得了巨大进展，但其有效性在很大程度上依赖于对正常数据分布的良好建模。缺乏足够的正常数据时，其准确检测异常的能力会显著降低。

CLIP 在大量图像-文本数据上训练，通过对比学习结合强大的语言模型和视觉特征编码器来捕捉强大的概念。这种组合使 CLIP 在图像分类中实现了令人印象深刻的零样本性能，因为它可以很好地推广到新类别，而无需特定任务的训练 [24–27, 42, 51]。最近，许多研究 [9, 12, 35, 39] 探索了将 CLIP 模型中嵌入的知识迁移到各种下游任务的方法，在图像描述、图像-文本检索和图像生成等领域取得了有希望的成果。这些努力展示了 CLIP 的多功能性及其推动发展的潜力

advancements across diverse applications.

Despite the rapid advancements achieved by CLIP, numerous studies have highlighted persistent limitations in the features it extracts. While CLIP demonstrates strong generalization across various tasks, it often struggles to capture nuanced details and essential spatial relationships, which are crucial for tasks demanding precise boundary delineation and fine-grained feature extraction. This limitation results in suboptimal performance in downstream applications, especially that require high levels of detail, such as object detection, scene segmentation, or tasks in medical imaging [14, 29, 30, 37, 49, 50, 57, 60]. As a result, leveraging CLIP for fine-granular tasks frequently necessitates task-specific adaptations to bridge the gap between its generalized feature extraction and the precision required for specialized applications.

CLIP-based Anomaly Detection There have been several efforts to leverage CLIP for AD tasks. One of the pioneering approaches, WinCLIP [19], proposes a method for extracting and aggregating visual features from multiple levels to align with text features, demonstrating the potential of CLIP in this context. Subsequent research investigates various adaptation methods to bridge the gap between natural domains and the AD domain, resulting in performance improvements. For instance, [7, 8, 18] focus on refining visual features by employing adapters to enhance patch-level visual representations. However, these approaches often rely on text embeddings from the original CLIP model as soft supervision and overlook a critical limitation of CLIP in AD: its unclearness in distinguishing between anomalous and normal semantics, particularly within the text encoder, resulting in suboptimal performance. Other works have employed prompt-learning-based methods[5, 6, 41, 59], introducing learnable embeddings into the text encoder to better represent abnormality. However, the class information in CLIP can be damaged, potentially degrading generalization, especially in data-limited and zero-shot settings.

Different from previous methods, we are the first to investigate CLIP's inherent limitation in capturing anomaly-aware information, specifically in differentiating between normal and anomalous semantics in text prompts. Rather than relying solely on the original anomaly-unaware text embeddings or unaltered feature spaces, our method is able to refine the embeddings to actively incorporate anomaly-discriminative representations.

3. Method

3.1. Overview

3.1.1. Problem Formulation

Zero-shot AD models are trained to identify anomalous samples whose categories may be unseen in the training dataset. Specifically, the model is expected to learn both normal and abnormal patterns that are shared across different classes given a training set $\mathcal{D}_{train}$ with normal or anomalous samples, in order to be capable of performing AD tasks on a series of different test datasets $\{\mathcal{D}^1_{test}, \mathcal{D}^2_{test}, ..., \mathcal{D}^n_{test}\}$, where each $\mathcal{D}^i_{test}$ is distinct from $\mathcal{D}_{train}$. Image-level AD can be formally defined as a binary classification problem, where the model aims to classify samples $x \in \mathcal{D}$ as either normal ($y = 0$) or anomalous ($y = 1$). Anomaly segmentation extends this concept to pixel-level with mask S, aiming to identify anomalous regions by highlighting pixels associated with anomalies.

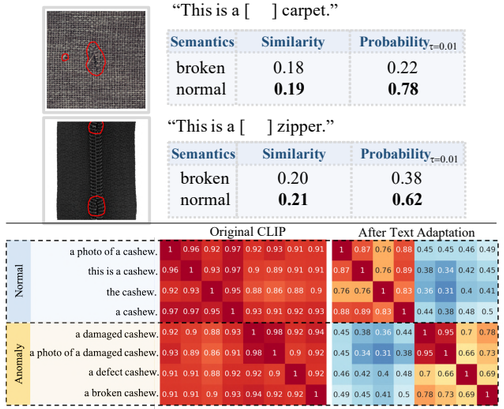


Figure 2. **(Top) Examples illustrating CLIP's Anomaly Unawareness.** Despite the obvious anomalies present in the images, image features have higher similarities to normal descriptions, rather than anomaly descriptions, mistakenly. This problem is enlarged with a low temperature τ. **(Bottom) Text Feature Similarity Heatmap among Normal and Anomaly Descriptions: Original CLIP vs. After Text Adaptation.** Red indicates high similarity. In original CLIP, normal features exhibit strong similarity with anomaly features, whereas text adaptation successfully separates them, clarifying the semantic distinctions between normal and anomaly descriptions.

3.1.2. Current Challenges

Anomaly Unawareness in CLIP: The CLIP-based AD method classifies visual features as "anomalies" if they exhibit greater similarities to anomaly prompt embeddings than to normal prompt embeddings, thus requiring well-defined boundaries between these two kinds of prompts. However, in real applications, CLIP's text embeddings often lack the clear separability needed to reliably distinguish between normal and anomaly classes.

We observe that, despite the visible defects in example images from the MVTec-AD [2], their features exhibit higher cosine similarity with "normal" prompts than with correct "anomaly" descriptions (see Fig. 2 (top)), indicating CLIP's inaccurate semantic understanding. Without adap-

在各类应用中的进展。

尽管CLIP取得了快速的进展，但许多研究指出它提取的特征仍存在持续的局限性。虽然CLIP在各种任务中表现出很强的泛化能力，但它经常难以捕捉细微的细节和关键的空间关系，而这些对于需要精确边界划分和精细特征提取的任务至关重要。这种局限性导致在下游应用中的性能不理想，尤其是在需要高细节水平的任务中，例如目标检测、场景分割或医学影像任务。因此，在精细任务中利用CLIP通常需要针对特定任务进行调整，以弥合其通用特征提取与专业应用所需精确性之间的差距。

基于CLIP的异常检测 在将CLIP用于异常检测（AD）任务方面已经有多项尝试。其中一个开创性的方法，WinCLIP [19], 提出了一种从多个层级提取和聚合视觉特征以与文本特征对齐的方法，展示了CLIP在该场景中的潜力。后续研究探索了各种适应方法，以弥合自然域与异常检测域之间的差距，从而提高性能。例如， [7, 8, 18] 专注于通过使用适配器提升补丁级视觉表示来优化视觉特征。然而，这些方法通常依赖于来自原始CLIP模型的文本嵌入作为软监督，并忽视了CLIP在异常检测中的一个关键局限性：其难以区分异常语义和正常语义，特别是在文本编码器中，导致性能不理想。其他工作采用了基于提示学习的方法 [5, 6, 41, 59], ，引入可学习的嵌入

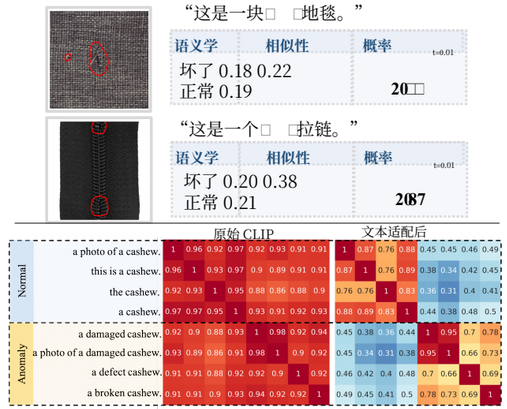


图 2.（上）示例说明 CLIP 对异常的无感知性。尽管图像中存在明显的异常，但图像特征与正常描述的相似度高于异常描述，从而导致错误判断。当温度较低时，这一问题会被放大 τ。（下）正常与异常描述之间的文本特征相似度热图：原始 CLIP 与文本适配后的对比。红色表示高相似度。在原始 CLIP 中，正常特征与异常特征表现出较强的相似性，而文本适配成功地将它们分开，明确了正常与异常描述之间的语义差异。

与以往的方法不同，我们是首个研究 CLIP 在捕捉异常感知信息方面固有局限性的工作，特别是在区分文本提示中的正常语义和异常语义方面。我们并不完全依赖原始的对异常不敏感的文本
在嵌入或未改变的特征空间中，我们的方法能够优化嵌入，以主动整合异常可区分的表示。

3. 方法

3.1. 概述

3.1.1. 问题的提出

零样本异常检测模型经过训练，以识别训练数据集中可能未出现过类别的异常样本。具体来说，期望模型学习在给定训练集 $\mathcal{D}_{train}$ （包含正常或异常样本）的情况下，共享于不同类别的正常和异常模式，以便能够在一系列不同的测试数据集 $\{\mathcal{D}^1_{test}, \mathcal{D}^2_{test}, ..., \mathcal{D}^n_{test}\}$ 上执行异常检测（AD）任务，其中每个 $\mathcal{D}^i_{test}$ 都与 $\mathcal{D}_{train}$ 不同。图像级异常检测可以正式定义为一个二分类问题，模型旨在将样本 $x \in \mathcal{D}$ 分类为正常 ($y = 0$) 或异常 ($y = 1$)。异常分割将这一概念扩展到像素级别，使用掩码 S，旨在通过突出与异常相关的像素来识别异常区域。

3.1.2. 当前挑战

CLIP中的异常无意识：基于CLIP的异常检测方法如果视觉特征与异常提示嵌入的相似度大于与正常提示嵌入的相似度，则将其分类为“异常”，因此需要在这两类提示之间有明确的界限。然而，在实际应用中，CLIP的文本嵌入往往缺乏可靠区分正常类和异常类所需的清晰可分性。

我们观察到，尽管来自 MVTec-AD [2]，的示例图像存在明显缺陷，但它们的特征与“正常”提示的余弦相似度比与正确的“异常”描述的相似度更高（见图 2（上）），这表明 CLIP 的语义理解不准确。没有

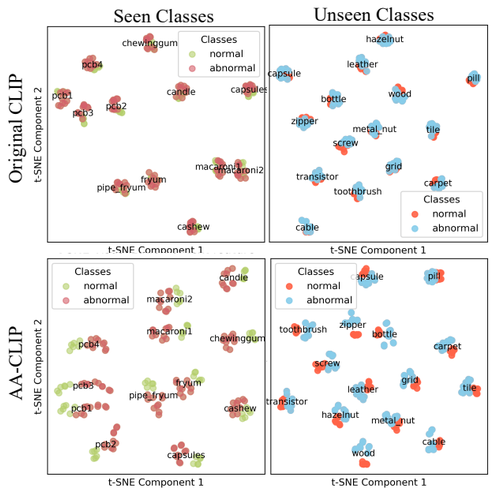


Figure 3. **t-SNE Visualization of Text Features from Original CLIP vs. AA-CLIP.** Each point represents a text feature encoded from a prompt. Original CLIP's normal and anomaly text features are intertwined, while our method effectively disentangles them. This disentanglement is generalizable to novel classes, validating the anomaly-awareness of our model.

tation, there persists a high similarity between the normal and abnormal text embeddings of a single class, as shown in Fig. 2 (bottom), suggesting a potential entanglement of normal and anomaly semantics within text space. We term this limitation **Anomaly Unawareness** and attribute it to the training process of CLIP: it is primarily trained on general, non-anomalous datasets and lacks specific guidance on defect detection. Consequently, it is challenging to rely on original CLIP embeddings to detect subtle or context-specific anomalies.

This issue remains evident across different categories in our t-SNE analysis: as shown in Fig. 3 (top), only subtle separations are observed within an object cluster, where text embeddings for both normal and abnormal semantics are intermixed. This entangled pattern may potentially lead to anomaly-unaware text-image alignment, which reinforces the necessity to *adapt CLIP's to enhance its ability of anomaly-awareness.*

Embedding Adaptation Dilemma: Discussion above renders the adaptation of CLIP essential for effective AD. However, since CLIP's embeddings are already optimized through extensive pretraining, it could be susceptible to overfitting to new dataset during adaptation. Overfitting convergence leads to minimized intra-class distinctions in the training data, often at the expense of the feature separability for effective generalization to unseen data.

To address this, *a carefully controlled refinement* is crucial to preserve CLIP's generalization capabilities while enhancing its sensitivity to anomalies.

3.1.3. Overview of Our Solution

Motivated by Sec. 3.1.2, we propose **Anomaly-Aware CLIP (AA-CLIP)** with improved anomaly awareness. As shown in Fig. 4, AA-CLIP is trained through a two-stage training strategy that sequentially adapts the semantic-rich text space and detail-focused visual space, with original CLIP parameters remaining frozen. In the first stage (see Fig. 4 (Top)), we incorporate Residual Adapters into the shallow layers of the text encoder, and the visual features from the fixed image encoder serve as a stable reference for optimization. A Disentangle Loss is purposed to enforce effective discrimination by ensuring independence between normal and anomaly embeddings. In the second stage, we integrate Residual Adapters into the shallow layers of the visual encoder to align patch-level features with the fixed, specially adapted texture features from the fixed text encoder (see in Fig. 4 (Bottom)). Ultimately, our AA-CLIP succeeds in equipping CLIP with anomaly awareness across seen and unseen classes, as shown in Fig. 3 (bottom).

3.2. AA-CLIP with Two-Stage Adaptation Strategy

3.2.1. Residual Adapter

To preserve CLIP's pre-trained knowledge while enabling targeted adaptation, we introduce lightweight Residual Adapters in the shallow layers (up to layer K) of both text and vision encoders.

The output feature $x^i \in \mathbb{R}^{N\times d}$ of CLIP's i-th ($i \leq K$) transformer layer is fed into the i-th adapter, outputting adapted feature $x^i_{residual}$, as shown in Eq. (1),

$$x^i_{residual} = Norm(Act(W^i x^i)), \tag{1}$$

where $W^i \in \mathbb{R}^{d\times d}$ is the trainable linear weight of i-th adapter, $Act(\cdot)$ is an activation function, and $Norm(\cdot)$ is a normalizing function. The original feature x^i and the enhanced feature $x^i_{residual}$ are fused in a weighted manner, generating $x^i_{enhanced}$, the input to the next transformer layer, as shown in Eq. (2),

$$x^i_{enhanced} = \lambda\, x^i_{residual} + (1-\lambda)\, x^i, \tag{2}$$

where λ is a hyper-parameter to control the residual ratio, adjusting the fusing degree of AD-specific knowledge for preserving the original CLIP's generalization ability and improved performance.

3.2.2. Two-Stage Training Strategy

Disentangling Anomaly-Aware Text Anchors: In the first stage, our objective is to learn anomaly-discriminative text anchors by adapting the text encoder while keeping the

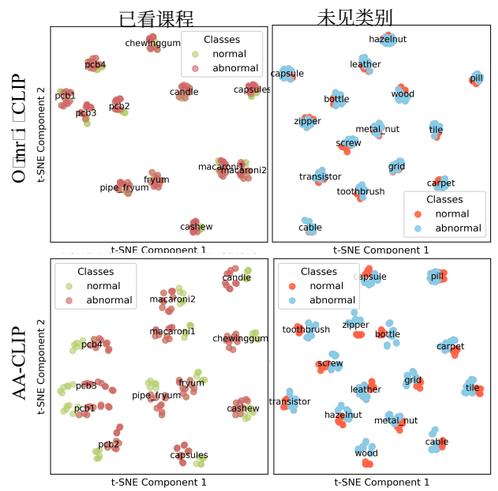


图 3. 原始 CLIP 与 AA-CLIP 文本特征的 t-SNE 可视化。每个点代表从提示编码的文本特征。原始 CLIP 的正常和异常文本特征交织在一起，而我们的方法有效地将它们解开。这种解耦可以推广到新类别，验证了我们模型的异常感知能力。

然而，不同情况下单一类别的正常与异常文本嵌入之间仍然存在高度相似性，如图2（底部）所示，这表明文本空间中正常语义与异常语义可能存在潜在的纠缠。我们将这一限制称为“异常无感知”，并将其归因于CLIP的训练过程：CLIP主要在通用的、非异常的数据集上进行训练，缺乏关于缺陷检测的具体指导。因此，依赖原始CLIP嵌入来检测细微或特定上下文的异常是具有挑战性的。

这个问题在我们对不同类别的 t-SNE 分析中仍然明显：如图 3（上）所示，在一个对象簇内仅观察到细微的分离，其中正常语义和异常语义的文本嵌入相互混合。这种纠缠的模式可能会导致对异常不敏感的文本-图像对齐，这强化了需要调整 CLIP 以增强其异常感知能力的必要性。

嵌入适应困境：上述讨论表明，为了实现有效的异常检测（AD），CLIP 的适应是必不可少的。然而，由于 CLIP 的嵌入已经通过大量预训练进行了优化，它在适应新数据集时可能容易过拟合。过拟合收敛会导致训练数据中类内差异最小化，通常以牺牲特征在未见数据上的有效可分性为代价。

为了解决这个问题，仔细控制的优化至关重要，以在增强其对异常的敏感性的同时保留CLIP的泛化能力。

3.1.3. 我们解决方案的概述

受第3.1.2节的启发，我们提出了具有改进异常感知能力的异常感知CLIP（AA-CLIP）。如图4所示，AA-CLIP通过两阶段训练策略进行训练，该策略依次适应语义丰富的文本空间和注重细节的视觉空间，同时保持原始CLIP参数冻结。在第一阶段（见图4（顶部）），我们在文本编码器的浅层中引入残差适配器，并且来自固定图像编码器的视觉特征作为优化的稳定参考。设定了解耦损失以通过确保正常和异常嵌入之间的独立性来实现有效的区分。在第二阶段，我们在视觉编码器的浅层中集成残差适配器，以将补丁级特征与来自固定文本编码器的固定、特意调整的纹理特征对齐（见图4（底部））。最终，我们的AA-CLIP成功装备了CLIP对已知和未知类别的异常感知能力，正如

3.2. 采用两阶段适应策略的AA-CLIP

3.2.1. 残差适配器

为了在保留 CLIP 预训练知识的同时实现有针对性的适应，我们在文本和视觉编码器的浅层（最多到第 K 层）引入了轻量级残差适配器。

CLIP第 i层($i \leq K$)变换器的输出特征 $x^i \in \mathbb{R}^{N\times d}$ 被输入到第 i个适配器，输出适配后的特征 $x^i_{residual}$，如公式(1)所示，

$$x^i_{residual} = Norm(Act(W^i x^i)), \tag{1}$$

$W^i \in \mathbb{R}^{d\times d}$ 是第 i个适配器的可训练线性权重，$Act(\cdot)$是激活函数，$Norm(\cdot)$是归一化函数。原始特征x^i和增强特征 $x^i_{residual}$ 以加权方式融合，生成 $x^i_{enhanced}$，作为下一变换器层的输入，如公式(2)所示，

$$x^i_{enhanced} = \lambda\, x^i_{residual} + (1-\lambda)\, x^i, \tag{2}$$

其中 λ 是一个超参数，用于控制残差比率，调整 AD 特定知识的融合程度，以保持原始 CLIP 的泛化能力并提升性能。

3.2.2. 两阶段训练策略

解开异常感知文本锚点：在第一阶段，我们的目标是通过调整文本编码器来学习异常判别的文本锚点，同时保持

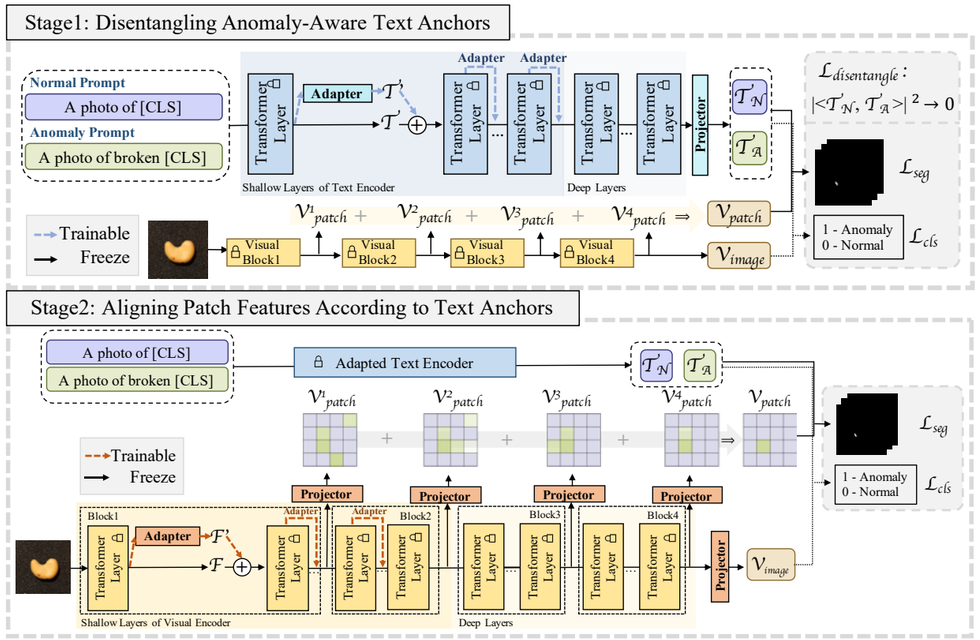


Figure 4. **The Two-Stage Training Pipeline of Anomaly-Aware CLIP.** In the first stage, the text encoder of AA-CLIP is trained to identify anomaly-related semantics, helped by a disentangle loss. In the second stage, patch features are aligned with these text anchors. Both stages are achieved by the integration of Residual Adapters into the shallow layers of CLIP's backbone. This controlled adaptation enables CLIP to effectively distinguish anomalies, which forms our Anomaly-Aware CLIP.

image encoder fixed. We incorporate Residual Adapters into the first K_T layers of the CLIP text encoder, as illustrated in Fig. 4 (Top), and set the final projector in the text encoder to be learnable to facilitate improved alignment.

Using prompts designed to encapsulate both normal and anomalous semantics (as detailed in Appendix), text encoder generates corresponding high-level embeddings. The average embeddings of the normal and anomaly prompts serve as our initial text anchors, denoted as T_N and $T_A \in \mathbb{R}^d$, respectively. These anchors are refined by being aligned with visual features extracted from an enhanced CLIP visual encoder, as [28, 59]. Alignment is conducted at both image and patch levels to incorporate both global and local semantics. By calculating the cosine similarity between these anchors and the image features $V_{image} \in \mathbb{R}^d$ or patch features $V_{patch} \in \mathbb{R}^{N\times d}$, as shown in Eq. (3),

$$\begin{aligned} p_{cls} &= CosSim(V_{image}, [T_N, T_A]), \\ p^o_{seg} &= CosSim(V_{patch}, [T_N, T_A]), \end{aligned} \tag{3}$$

where $[\cdot,\cdot]$ means concatenate operation, we obtain the classification prediction $p_{cls} \in \mathbb{R}^2$ and the segmentation prediction $p^o_{seg} \in \mathbb{R}^{N\times 2}$. The segmentation prediction p^o_{seg} is then reshaped and upsampled to $p_{seg} \in \mathbb{R}^{H\times W\times 2}$ to align with the height H and width W of segmentation mask S. Following previous works [6, 7, 18, 59], we compute the classification loss $\mathcal{L}_{cls}$ and segmentation loss $\mathcal{L}_{seg}$ to optimize parameters, as specified in Eq. (4). Specifically, the classification loss is a binary cross-entropy that compares classification predictions with ground-truth labels y, and the segmentation loss is a combination of dice loss and focal loss applied to segmentation predictions and the anomaly segmentation mask S.

$$\begin{aligned} \mathcal{L}_{cls} &= \text{BCE}(p_{cls}, y), \\ \mathcal{L}_{seg} &= \text{Dice}(p_{seg}, S) + \text{Focal}(p_{seg}, S), \\ \mathcal{L}_{align} &= \mathcal{L}_{cls} + \mathcal{L}_{seg}. \end{aligned} \tag{4}$$

To enhance the separation between normal and anomaly text embeddings, we introduce a Disentangle Loss encouraging orthogonality between T_N and T_A to minimize correlation, as in Eq. (5):

$$\mathcal{L}_{dis} = |< T_N, T_A >|^2. \tag{5}$$

The Disentangle Loss $\mathcal{L}_{dis}$ is incorporated into the alignment loss $\mathcal{L}_{align}$ as a regularization term, weighted by a factor γ, which forms the total loss, as in Eq. (6):

$$\mathcal{L}_{total} = \mathcal{L}_{align} + \gamma\mathcal{L}_{dis}. \tag{6}$$

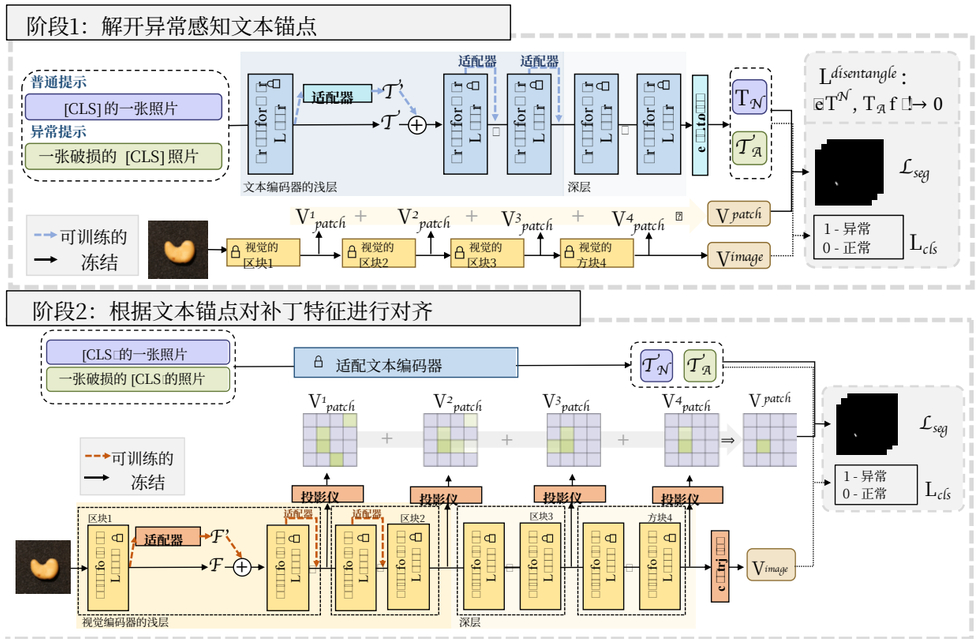


图 4. 异常感知 CLIP 的两阶段训练流程。在第一阶段，AA-CLIP 的文本编码器通过解耦损失的帮助，被训练以识别与异常相关的语义。在第二阶段，图像块特征与这些文本锚点进行对齐。两个阶段都是通过将残差适配器集成到 CLIP 主干的浅层实现的。这种受控适配使 CLIP 能够有效区分异常，从而形成我们的异常感知 CLIP。

图像编码器已修复。我们将残差适配器（Residual Adapters）加入到 CLIP 文本编码器的前 K_T 层，如图 4（上）所示，并将文本编码器中的最终投影器设置为可学习，以便促进更好的对齐。

使用旨在囊括正常和异常语义的提示（详见附录），文本编码器生成相应的高层嵌入。正常和异常提示的平均嵌入作为我们的初始文本锚点，分别表示为 T_N 和 $T_A \in \mathbb{R}^d$。这些锚点通过与从增强的 CLIP 视觉编码器中提取的视觉特征对齐而得到优化，表示为 [28, 59]。对齐在图像级和补丁级进行，以整合全局和局部语义。通过计算这些锚点与图像特征 $V_{image} \in \mathbb{R}^d$ 或补丁特征 $V_{patch} \in \mathbb{R}^{N\times d}$ 之间的余弦相似度，如公式 (3) 所示，

$$\begin{aligned} p_{cls} &= CosSim(V_{image}, [T_N, T_A]), \\ p^o_{seg} &= CosSim(V_{patch}, [T_N, T_A]), \end{aligned} \tag{3}$$

其中 $[\cdot,\cdot]$ 表示连接操作，我们得到分类预测 $p_{cls} \in \mathbb{R}^2$ 和分割预测 $p^o_{seg} \in \mathbb{R}^{N\times 2}$。然后将分割预测 p^o_{seg} 重塑并上采样到 $p_{seg} \in \mathbb{R}^{H\times W\times 2}$ 以进行对齐使用分割掩码 S 的高度 H 和宽度 W。遵循之前的工作 [6, 7, 18, 59], ，我们计算分类损失 $\mathcal{L}_{cls}$ 和分割损失 $\mathcal{L}_{seg}$ 来优化参数，如公式 (4) 所示。具体而言，分类损失是将分类预测与真实标签 y 进行比较的二元交叉熵，分割损失则是应用于分割预测和异常分割掩码 S 的 dice 损失与 focal 损失的组合。

$$\begin{aligned} \mathcal{L}_{cls} &= \text{BCE}(p_{cls}, y), \\ \mathcal{L}_{seg} &= \text{Dice}(p_{seg}, S) + \text{Focal}(p_{seg}, S), \\ \mathcal{L}_{align} &= \mathcal{L}_{cls} + \mathcal{L}_{seg}. \end{aligned} \tag{4}$$

为了增强正常文本和异常文本嵌入之间的区分，我们引入了一种解耦损失，鼓励 T_N 和 T_A 之间的正交性，以最小化相关性，如公式 (5) 所示：

$$\mathcal{L}_{dis} = |< T_N, T_A >|^2. \tag{5}$$

解缠损失 $\mathcal{L}_{dis}$ 被作为正则化项纳入对齐损失 $\mathcal{L}_{align}$ 中，并由因子 γ 加权，形成总损失，如公式 (6) 所示：

$$\mathcal{L}_{total} = \mathcal{L}_{align} + \gamma\mathcal{L}_{dis}. \tag{6}$$

In this stage, the distinction between normal and anomaly semantics is embedded into CLIP's text encoder while its original object-recognition capability is preserved. Figure 3 indicates that this ability of anomaly-awareness is robust and generalizable to novel classes.

Aligning Patch Features According to Text Anchors: Anomaly-aware semantic anchors can facilitate the adaptation of patch features, thereby improving the effectiveness and generalizability of anomaly localization. To achieve alignment between patch features and anchors from the previous stage, we introduce trainable Residual Adapters within the initial K_I layers of the CLIP visual encoder.

Features with multi-granularities are utilized to enhance segmentation [7, 19, 59]. Specifically, as shown in Fig. 4 (bottom), the intermediate output feature F^i are extracted from four distinct granularities. These multi-granularity features are then projected to align with the channel of text anchors via a trainable projector $Proj_i(\cdot)$, yielding V^i_{patch} at four distinct levels of granularity. The aggregated output V_{patch} is computed by summing individual V^i_{patch} outputs, as in Eq. (7):

$$\begin{aligned} V^i_{patch} &= Proj_i(F^i),\ i \in \{1,2,3,4\} \\ V_{patch} &= \sum_{i=1}^{4} V^i_{patch}. \end{aligned} \tag{7}$$

The cosine similarity scores between the aggregated V_{patch} and the text anchors are calculated to generate patch-level predictions as Eq. (3), resulting in the prediction maps.

During training, alignment is guided by the loss function defined in Eq. (4), facilitating both global and local alignment. During inference, anomaly prediction maps and corresponding anomaly scores are derived by comparing the similarity scores of visual features against normal and anomaly text embeddings.

4. Experiments

4.1. Experiment Setups

Datasets We evaluate our model on 11 widely used benchmarks, as previous AD works [6, 7, 18, 19, 59], with distinct foreground objects spanning a variety of modalities, including photography, endoscopy, CT, MRI, and OCT. For the industrial domain, we use MVTec AD [2], VisA [62], BTAD [34] and MPDD [20]. For medical domain, we use brain MRI, liver CT and retina OCT from BMAD [1], and four different colon polyp detection datasets with different views (CVC-ClinicDB [4], CVC-ColonDB [3], Kvasir-SEG [40] and CVC-300 [47]). Each dataset has both image-level labels and pixel-level masks for evaluation.

We train our model on a real-world industrial AD dataset - VisA [62] - in which objects are different from other datasets. Results of VisA are obtained using MVTec-AD as the training dataset. To demonstrate adaptation efficiency, we conduct training under various data levels: 2-shot per class, 16-shot per class, 64-shot per class, and full-shot. The corresponding number of samples are randomly selected from each class, while maintaining a consistent 1:1 ratio between normal and anomaly samples.

Metrics Following [6, 7, 10, 11, 18, 19, 43, 55], we use the Area Under the Receiver Operating Characteristic Curve (AUROC) as the metric. We compute AUROC at both the image and pixel levels to comprehensively assess the model's effectiveness in detecting and localizing anomalies.

Implementation Details Following [6, 7, 41, 59], we use OpenCLIP with the ViT-L/14 architecture as the backbone, and input images are resized to 518×518. All parameters of CLIP remain frozen. We set λ to 0.1, K_T to 3, K_I to 6, and γ to 0.1. For multi-level feature extraction, we utilize outputs from the 6-th, 12-th, 18-th, and 24-th layers of the visual encoder to compose the overall output. For the first stage, we train the model for 5 epochs with a learning rate of 1×10^{-5}. For the second stage, we continue training for 20 epochs, adjusting the learning rate to 5×10^{-4}. Parameters are updated by Adam optimizers. All experiments are conducted on a single NVIDIA GeForce RTX 3090 GPU. More details are available in Appendix.

4.2. Comparison with SOTA Methods

We compare our method against CLIP and several recent SOTA models. Among them, WinCLIP [19], VAND [7] and MVFA-AD [18] use original CLIP text encoder, and AnomalyCLIP [59] and AdaCLIP [6] incorporate learnable prompts. To ensure a fair comparison, we re-train models that are originally trained on different datasets to match the dataset settings of other approaches (detailed in Appendix).

Quantitative results are presented in Tab. 1 and Tab. 2. Although adapting only the patch feature with original text embeddings has made progress in AD, the superior performance of AA-CLIP highlights its effective disentanglement of anomaly-discriminative semantics, leading to further progress. Notably, even in data-limited situations, our method consistently demonstrates top performance. At the pixel level, with only 2 shots per class used for training, our method achieves improved average zero-shot performance compared to previous methods. With the full dataset, we set a new pixel-level SOTA with an AUROC of 93.4%. At the image level, our method is competitive with just 2 shots for training and establishes a new SOTA of 83.1% with 64 shots per class.

Unlike previous methods, our approach does not rely heavily on data resources to achieve top-tier performance. Comparison under different levels of data available, as shown in Fig. 5, reveals that our approach consistently outperforms other methods in general. Even with limited data, our model reaches competitive results, while other methods

在这个阶段，正常与异常语义的区分被嵌入到CLIP 的文本编码器中，同时保留了其原有的物体识别能力。图 3 表明，这种异常感知能力具有鲁棒性，并且可以推广到新类别。

根据文本锚点对补丁特征进行对齐：异常感知语义锚点可以促进补丁特征的适应，从而提高异常定位的有效性和泛化能力。为了实现补丁特征与上一阶段锚点之间的对齐，我们在CLIP视觉编码器的初始 K_I 层中引入可训练的残差适配器。

利用多粒度特征来增强分割 [7, 19, 59]。具体来说，如图 4（下部）所示，从四个不同粒度提取中间输出特征 F^i 。这些多粒度特征随后通过可训练投影器 $Proj_i(\cdot)$投影，以与文本锚点的通道对齐，从而在四个不同粒度级别生成 V^i_{patch}。聚合输出 V_{patch} 是通过对各个 V^i_{patch} 输出求和计算的，如公式 (7) 所示：

$$\begin{aligned} V^i_{patch} &= Proj_i(F^i),\ i \in \{1,2,3,4\} \\ V_{patch} &= \sum_{i=1}^{4} V^i_{patch}. \end{aligned} \tag{7}$$

计算聚合的 V_{patch}与文本锚点之间的余弦相似度得分，以生成如公式(3)所示的补丁级预测，从而得到预测图。

在训练过程中，对齐由式(4)定义的损失函数指导，促进全局和局部对齐。在推理过程中，通过将视觉特征的相似性分数与正常和异常文本嵌入进行比较，得到异常预测图和相应的异常分数。

4. 实验

4.1. 实验设置

数据集 我们在 11 个广泛使用的基准上评估我们的模型，正如之前的异常检测（AD）工作 [6, 7, 18, 19, 59], , 这些基准具有不同的前景对象，涵盖多种模态，包括摄影、内窥镜、CT、MRI 和 OCT。对于工业领域，我们使用 MVTec AD [2], 、VisA [62],、BTAD [34] 和 MPDD [20]。对于医疗领域，我们使用 BMAD [1], 的脑 MRI、肝脏 CT 和视网膜 OCT，以及四个不同结肠息肉检测数据集，这些数据集具有不同视角（CVC-ClinicDB [4], 、CVC-ColonDB [3], 、Kvasir-SEG [40] 和 CVC-300 [47]）。每个数据集都具有用于评估的图像级标签和像素级掩码。

我们在一个真实世界的工业异常检测数据集 - VisA [62] 上训练我们的模型，其中的对象与其他数据集不同。VisA的结果是使用MVTec-AD获得的

训练数据集。为了展示适应效率，我们在不同的数据量下进行训练：每类2个样本、每类16个样本、每类64个样本，以及全样本。每类的对应样本数量是随机选择的，同时保持正常样本与异常样本之间的1:1比例一致。

指标 以下 [6, 7, 10, 11, 18, 19, 43, 55], 我们使用接收器操作特征曲线下面积（AUROC）作为指标。我们在图像和像素两个层面计算 AUROC，以全面评估模型在检测和定位异常方面的有效性。

实现细节 参照 [6, 7, 41, 59], , 我们使用 ViT-L/14 架构的 OpenCLIP 作为主干，输入图像被调整为 518×518。CLIP 的所有参数保持冻结。我们将 λ 设置为 0.1， K_T 设置为 3， K_I 设置为 6， γ 设置为 0.1。为了进行多层特征提取，我们利用视觉编码器第 6、12、18 和 24 层的输出组成整体输出。在第一阶段，我们以学习率 1×10^{-5} 训练模型 5 个 epoch。在第二阶段，我们继续训练 20 个 epoch，并将学习率调整为 5×10^{-4}。参数通过 Adam 优化器更新。所有实验均在单个 NVIDIA GeForce RTX 3090 GPU 上进行。更多细节见附录。

4.2. 与SOTA方法的比较

我们将我们的方法与 CLIP 以及几种最近的 SOTA 模型进行了比较。其中，WinCLIP [19], VAND [7] 和 MVFA-AD [18] 使用了原始的 CLIP 文本编码器，而 AnomalyCLIP [59] 和 AdaCLIP [6] 则引入了可学习的提示。为确保公平比较，我们对那些最初在不同数据集上训练的模型进行了重新训练，以匹配其他方法的数据集设置（详细信息见附录）。

定量结果如表 1 和表 2 所示。尽管仅使用原始文本嵌入适配补丁特征在异常检测（AD）方面取得了一定进展，AA-CLIP 的优越性能仍凸显了其对异常判别语义的有效解耦，从而推动了进一步的进展。值得注意的是，即使在数据有限的情况下，我们的方法仍能持续展示顶尖性能。在像素级别上，仅使用每类 2 个样本进行训练，我们的方法相比以往方法实现了更好的平均零样本性能。在使用完整数据集时，我们在像素级别上实现了新的 SOTA，AUROC 达到 93.4%。在图像级别上，我们的方法在仅使用每类 2 个样本训练时仍具竞争力，并在每类 64 个样本训练时建立了新的 SOTA，达到 83.1%。

与以往的方法不同，我们的方法并不依赖大量数据资源也能实现顶级性能。如图5所示，在不同数据可用水平下的比较显示，我们的方法总体上始终优于其他方法。即使在数据有限的情况下，我们的模型也能达到有竞争力的结果，而其他方法则无法做到。

Domain	Dataset	CLIP*	WinCLIP*	VAND*	MVFA-AD	AnomalyCLIP*	AdaCLIP	**Ours**			
		OpenCLIP	CVPR 2023	CVPRw 2023	CVPR 2024	ICLR2024	ECCV2024	-			
	Available training shots	-	-	full	full	full	full	2	16	64	full
Industrial	BTAD	30.6	32.8	91.1	90.1	93.3	90.8	92.8	94.4	96.5	**97.0**
	MPDD	62.1	95.2	94.9	94.5	96.2	96.6	96.3	96.5	96.3	**96.7**
	MVTec-AD	38.4	85.1	87.6	84.9	91.1	89.9	91.0	91.2	91.6	**91.9**
	VisA	46.6	79.6	94.2	93.4	95.4	**95.5**	93.4	93.8	94.0	**95.5**
Medical	Brain MRI	68.3	86.0	94.5	95.6	96.2	93.9	96.3	96.4	**96.5**	95.5
	Liver CT	90.5	96.2	95.6	96.8	93.9	94.5	97.3	97.7	97.7	**97.8**
	Retina OCT	21.3	80.6	88.5	90.9	92.6	88.5	94.2	95.1	94.4	**95.5**
	ColonDB	49.5	51.2	78.2	78.4	82.9	80.0	83.9	83.5	**84.7**	84.0
	ClinicDB	47.5	70.3	85.1	83.9	85.0	85.9	89.2	87.6	87.8	**89.9**
	Kvasir	44.6	69.7	80.3	81.9	81.9	86.4	82.1	84.6	85.2	**87.2**
	CVC-300	49.9	-	92.8	82.6	95.4	92.9	96.0	**97.4**	96.0	96.4
	Average	49.9	74.7	89.3	88.5	91.3	90.4	92.0	92.6	92.8	**93.4**

Table 1. Pixel-level AUROC of zero-shot AD methods in Industrial and Medical domains. Method sources and the number of shots used for training are noted. Results of methods with * are copied from the papers or inferred from official weight. Best results are highlighted as **first**, second and third.

Domain	Dataset	CLIP&VAND*	WinCLIP*	MVFA-AD	AnomalyCLIP*	AdaCLIP	**Ours**			
		OpenCLIP	CVPR 2023	CVPR 2024	ICLR2024	ECCV2024	-			
	Available training shots	-	-	full	full	full	2	16	64	full
Industrial	BTAD	73.6	68.2	94.3	85.3	90.9	88.0	90.9	94.7	**94.8**
	MPDD	73.0	63.6	70.9	73.7	72.1	63.6	**78.3**	75.7	75.1
	MVTec-AD	86.1	91.8	86.6	90.9	90.0	85.9	89.7	**92.0**	90.5
	VisA	66.4	78.0	76.5	82.1	84.3	78.4	84.0	84.1	**84.6**
Medical	Brain MRI	58.8	66.5	70.9	83.3	80.2	**84.3**	80.4	83.4	80.2
	Liver CT	54.7	64.2	63.0	61.6	64.2	69.4	68.1	69.2	**69.7**
	Retina OCT	65.6	42.5	77.3	75.7	82.7	77.4	81.0	**82.9**	82.7
	Average	68.3	67.8	77.1	78.4	80.6	78.1	81.8	**83.1**	82.5

Table 2. Image-level AUROC of zero-shot AD methods in Industrial and Medical domains. Method sources and the number of shots used for training are noted. Results of methods with * are copied from the papers or inferred from official weight. Best results are highlighted as **first**, second and third.

display signs of underfitting. As data increases, our method maintains its lead, establishing a new SOTA at both pixel and image levels.

4.3. Visualization

To illustrate the alignment intuitively, we present visualization examples in Fig. 6 with original configuration for previous works. Although previous methods with can detect anomalous regions, our AA-CLIP demonstrates fewer false-negative predictions in both industrial and medical domains, accurately highlighting the correct anomaly regions.

4.4. Ablations Analysis

We conduct thorough ablation experiments of our refinement of both visual and text space, as shown in Tab. 3 and Fig. 7. The second row in Tab. 3, which mirrors the structure of VAND [7], serves as our baseline.

Image Space: As shown in Tab. 3 line "2.", inserting the vallina linear adapter into transformer layers results in a significant decline in zero-shot performance, indicating the damage of the original generalization ability of CLIP. Incorporating our Residual Adapters mitigates this issue (shown in line "3."), enhancing performance while preserving original information stored in CLIP.

Text Space: The last two rows in Tab. 3 highlight the impact of our approach in equipping CLIP's encoder with

	Method	Avg. AUROC	
		Pixel-Level	Image-Level
	CLIP	50.3	69.3
Image	1. + Linear Proj. *(VAND [7])*	88.9	69.3
	2. + Adapter	48.9(-40.0)	53.4(-15.9)
	3. **+ Residual Adapter**	91.3(+2.4)	80.7(+11.4)
Text	4. **+ Residual Adapter**	92.1(+3.2)	82.6(+13.3)
	5. **+ Disentangle Loss**	92.7(+3.8)	83.3(+14.0)

Table 3. **Ablation Study of Our Training Strategy with VisA-Trained 64-Shot Setup.** Our contributions are **bold**. While VAND uses linear projectors to improve AD performance, incorporating Residual Adapters further refines patch feature adaptation. Moreover, integrating our Disentangle Loss yields the best overall results.

域	数据集	CLIP*	WinCLIP*	VAND*	MVFA-AD 异常CLIP*	AdaCLIP	**Ours**			
		OpenCLIP	CVPR 2023	CVPRw 2023	CVPR 2024	ICLR2024	ECCV2024	-			
	可用训练射击	-	-	full	full	full	full	2	16	64	full
工业的	BTAD	30.6	32.8	91.1	90.1	93.3	90.8	92.8	94.4	96.5	**97.0**
	MPDD	62.1	95.2	94.9	94.5	96.2	96.6	96.3	96.5	96.3	**96.7**
	MVTec-AD	38.4	85.1	87.6	84.9	91.1	89.9	91.0	91.2	91.6	**91.9**
	VisA	46.6	79.6	94.2	93.4	95.4	**95.5**	93.4	93.8	94.0	**95.5**
医学	脑部核磁共振	68.3	86.0	94.5	95.6	96.2	93.9	96.3	96.4	**96.5**	95.5
	肝脏CT	90.5	96.2	95.6	96.8	93.9	94.5	97.3	97.7	97.7	**97.8**
	视网膜光学相干断层扫描	21.3	80.6	88.5	90.9	92.6	88.5	94.2	95.1	94.4	**95.5**
	ColonDB	49.5	51.2	78.2	78.4	82.9	80.0	83.9	83.5	**84.7**	84.0
	诊所数据库	47.5	70.3	85.1	83.9	85.0	85.9	89.2	87.6	87.8	**89.9**
	克瓦西尔	44.6	69.7	80.3	81.9	81.9	86.4	82.1	84.6	85.2	**87.2**
	CVC-300	49.9	-	92.8	82.6	95.4	92.9	96.0	**97.4**	96.0	96.4
	平均	49.9	74.7	89.3	88.5	91.3	90.4	92.0	92.6	92.8	**93.4**

表 1. 工业和医疗领域零样本异常检测方法的像素级 AUROC。标注了方法来源和训练使用的样本数。带有 * 的方法结果摘自论文或根据官方权重推断。最佳结果以第一、第二和第三名突出显示。

域	数据集	CLIP&VAND*	WinCLIP*	MVFA-AD 异常CLIP*	AdaCLIP	**Ours**			
		OpenCLIP	CVPR 2023	CVPR 2024	ICLR2024	ECCV2024	-			
	可用训练射击	-	-	full	full	full	2	16	64	full
工业的	BTAD	73.6	68.2	94.3	85.3	90.9	88.0	90.9	94.7	**94.8**
	MPDD	73.0	63.6	70.9	73.7	72.1	63.6	**78.3**	75.7	75.1
	MVTec-AD	86.1	91.8	86.6	90.9	90.0	85.9	89.7	**92.0**	90.5
	VisA	66.4	78.0	76.5	82.1	84.3	78.4	84.0	84.1	**84.6**
医学	脑部核磁共振	58.8	66.5	70.9	83.3	80.2	**84.3**	80.4	83.4	80.2
	肝脏CT	54.7	64.2	63.0	61.6	64.2	69.4	68.1	69.2	**69.7**
	视网膜光学相干断层扫描	65.6	42.5	77.3	75.7	82.7	77.4	81.0	**82.9**	82.7
	平均	68.3	67.8	77.1	78.4	80.6	78.1	81.8	**83.1**	82.5

表2. 工业和医疗领域零样本异常检测方法的图像级AUROC。方法来源以及用于训练的样本数量已注明。带有 * 的方法结果来自论文或从官方权重推断。最佳结果以第一、第二和第三名突出显示。

显示出欠拟合的迹象。随着数据的增加，我们的方法保持领先，在像素级和图像级都建立了新的最先进水平（SOTA）。

4.3. 可视化

为了直观地说明对齐，我们在图6中展示了使用之前工作的原始配置的可视化示例。虽然之前的方法可以检测异常区域，但我们的AA-CLIP在工业和医疗领域中表现出更少的假阴性预测，能够准确突显正确的异常区域。

4.4. 消融分析

我们对视觉空间和文本空间的改进进行了彻底的消融实验，如表 3 和图 7 所示。表 3 的第二行，其结构与 VAND [7]，相同，作为我们的基线。

图像空间：如表3第"2."行所示，将原始化线性适配器插入变压器层会导致零样本性能显著下降，这表明

	方法	平均AUROC	
		像素级	图像级
	CLIP	50.3	69.3
图像	1. +线性投影 (VAND *[7]*)	88.9	69.3
	2. + 适配器	48.9(-40.0)	53.4(-15.9)
	3. **+ 残差适配器**	91.3(+2.4)	80.7(+11.4)
Text	4. **+ 残差适配器**	92.1(+3.2)	82.6(+13.3)
	5. **+ 解缠损失**	92.7(+3.8)	83.3(+14.0)

表3. 我们在 VisA 训练的 64-Shot 设置下的训练策略消融研究。我们的贡献以加粗显示。虽然 VAND 使用线性映射器来提升 AD 性能，但加入残差适配器进一步优化了补丁特征的适配。此外，整合我们的解耦损失能够得到最佳的整体结果。

对 CLIP 原始泛化能力的损害。引入我们的残差适配器可以缓解这个问题（见第"3."行），在保留 CLIP 中存储的原始信息的同时提升性能。

文本空间：表3的最后两行突出了我们的方法在赋予 CLIP编码器能力方面的影响

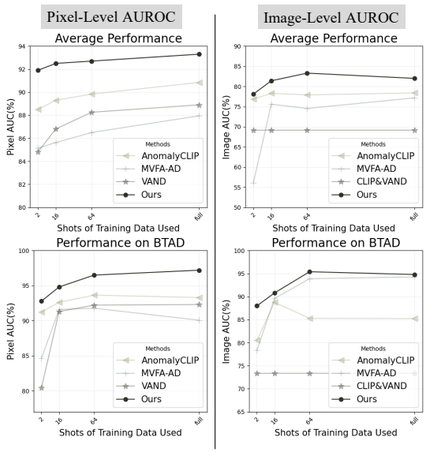


Figure 5. **Average Results (Top) and Results on BTAD (Bottom) of Different methods Trained on 2-, 16-, 64-shot per Class and Full Data of VisA.** Our method shows high fitting efficiency, achieving strong results across all data scales.

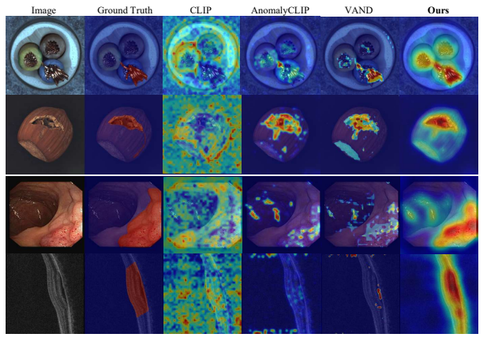


Figure 6. **Visualization of Anomaly Localization Results of Original CLIP [42], AnomalyCLIP [59], VAND [7] and our AA-CLIP.** Compared to previous methods, AA-CLIP demonstrates more reliable prediction capabilities in localizing anomaly.

anomaly-aware semantics. Line "4." validates that, with AA-CLIP, the model's ability to discriminate anomalies further improves, as the AA-CLIP's text encoder provides a more precise semantic foundation. Adding Disentangle Loss leads to an additional improvement (shown in Line "5."), especially at image-level, validating the necessity of independence between normal and anomaly anchors. These results underscore the crucial role of text space refinement in improved anomaly localization and classification.

Two-Stage Training: To validate the necessity of two-stage training, we adapt both text and image encoders together within one stage (also adopted by AdaCLIP). As shown

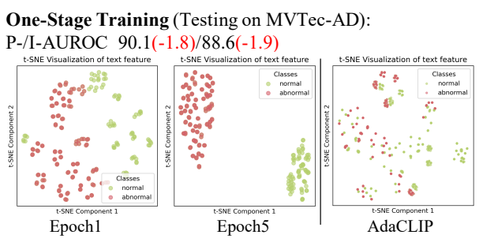


Figure 7. **Visualization of Text Space from One-Stage Training and from AdaCLIP.** During one-stage training, class information collapses easily, leading to damaged zero-shot performance.

in Fig. 7, one-stage model can easily exaggerate anomaly semantics and forget class information embedded in CLIP, damaging the model's generalization ability. The two-stage training strategy allows controlled adaptation, preserving CLIP's class-relevant knowledge in one end while adapting the other, as shown in Fig. 3.

5. Conclusion and Discussion

To our knowledge, this is the first work to explicitly analyze the intrinsic Anomaly Unawareness problem in CLIP. To tackle this issue, we propose a simple yet effective two-stage training strategy to embed anomaly-aware information into CLIP, enabling clear disentanglement of anomaly representations across both seen and novel classes. By leveraging residual adapters, our method preserves CLIP's strong generalization ability, achieving outstanding zero-shot performance across multiple datasets.

Our adapted AA-CLIP, developed through this two-stage adaptation strategy, reveals the potential of refining CLIP's feature space for improved performance in downstream applications. Beyond addressing anomaly unawareness, our work also provides a potential foundation for tackling other "unawareness" issues within CLIP. These may include limitations in context-awareness or specificity to domain-relevant nuances, suggesting further applications of our method in expanding CLIP's adaptability across diverse tasks. Additionally, we observe signs of overfitting with full-shot training, suggesting potential saturation during CLIP adaptation and warranting further investigation.

Acknowledgement

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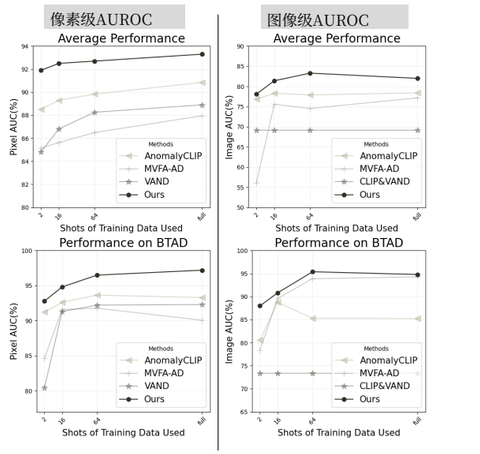


图5. 不同方法在VisA的2、16、64次样本和全部数据上训练的平均结果（上）及在BTAD上的结果（下）。我们的方法显示出高拟合效率，在所有数据规模上都取得了强劲的结果。

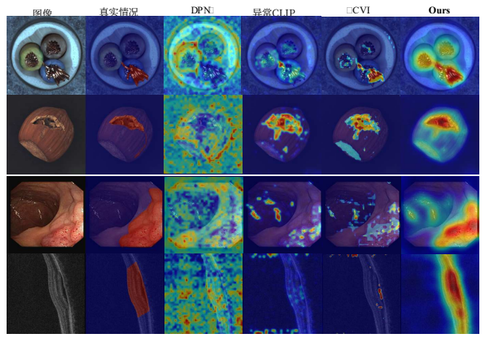


图6. 原始CLIP [42]、AnomalyCLIP [59]、VAND [7]以及我们的AA-CLIP的异常定位结果可视化。与之前的方法相比，AA-CLIP在定位异常方面展示了更可靠的预测能力。

异常感知语义。第"4"行验证了，使用 AA-CLIP 时，模型区分异常的能力进一步提升，因为 AA-CLIP 的文本编码器提供了更精确的语义基础。加入解耦损失会带来额外的提升（如第"5"行所示），尤其是在图像层面，验证了正常锚点与异常锚点独立性的必要性。这些结果强调了文本空间优化在提升异常定位和分类中的关键作用。

两阶段训练：为了验证两阶段训练的必要性，我们在一个阶段中同时调整文本和图像编码器（AdaCLIP 也采用了这种做法）。如图所示

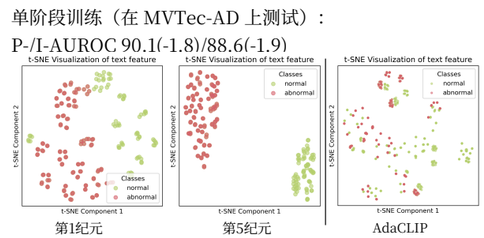


图7. 一阶段训练和AdaCLIP的文本空间可视化。在一阶段训练中，类别信息容易崩塌，导致零样本性能受损。

在图7中，单阶段模型容易夸大异常语义并忘记嵌入在CLIP中的类别信息，从而损害模型的泛化能力。两阶段训练策略允许受控适应，在一端保留CLIP的类别相关知识，同时在另一端进行调整，如图3所示。

5. 结论与讨论

据我们所知，这是首个明确分析CLIP中内在异常无意识问题的工作。为解决该问题，我们提出了一种简单而有效的两阶段训练策略，将异常感知信息嵌入CLIP，使得在已见类别和新类别中能够清晰地解耦异常表示。通过利用残差适配器，我们的方法保留了CLIP的强大泛化能力，在多个数据集上实现了出色的零样本性能。

我们通过这一两阶段适应策略开发的改良版AA-CLIP 展示了在下游应用中通过优化 CLIP 的特征空间以提升性能的潜力。除了应对异常认知不足外，我们的工作还为解决 CLIP 中其他"认知不足"问题提供了潜在基础。这可能包括对上下文感知的局限性或对领域相关细微差异的特异性，从而表明我们的方法在扩大CLIP 在不同任务中的适应性方面可能有更多应用。此外，我们在全样本训练中观察到过拟合的迹象，这表明CLIP 适应过程中可能存在饱和，需要进一步研究。

致谢

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